Humoyun Turaev (Ray) · 후모윤

Senior Lead Developer, Medical Ordering Platform · Full-Stack & DevOps

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SUMMARY

Senior full-stack and DevOps engineer with 5+ years of experience. As Senior Lead Developer for the medical ordering platform at Seeann Solution (Songdo, Incheon), Medical Software R&D Team, I'm responsible for the company's web services, their servers and databases: 7 services, legacy included, on 3 AWS servers and an on-prem GPU server. I built the internal office platform (in production) and the next-generation ordering platform (being prepared for production). I also handle scaling, bottleneck analysis, cost optimization and AI automation for work and operations. I start from what can fail, and I measure before and after every change.

EXPERIENCE

Seeann Solution Songdo, Incheon Apr 2026 – Present

Senior Lead Developer, Medical Ordering Platform · Medical Software R&D Team

- Rebuilding 3 legacy apps as one Next.js 16 / NestJS platform behind the same API contract, so the external desktop planning software keeps working unchanged (491 commits, 28 migrations, production deploy in preparation).
- Server-issued order codes: a per-day sequence row locked FOR UPDATE and committed with an idempotency key. Load test with 200 doctors ordering at once on 2 instances: **2.3 s, 0 mix-ups**.
- Zero browser API calls on cold load (SSR prefetch + hydration) and ETag/304 caching. Repeat-visit TTFB **82 → 32 ms**, list API **291 → 966 rps**.
- GPU job queue on MySQL 5.7 with leases and fencing tokens. Legacy AI pipeline peak memory **1.68 GiB → 122 MiB**, **205 → 145 s** per case.
- S3 as the system of record with an outbox and delete-only-after-proof cleanup. Queue wait behind a large file **34 → 1.5 s** (legacy).
- Operations: removed the causes of **1,800+** PM2 restarts, cut DB pools from up to 1,000 to 50 per app (limit 151), automated SSL for 15 hostnames, daily DB backups kept 30 days.
- Rolling out an AI-assisted self-healing ops agent (since Oct 2026): health checks on PM2, Docker, NGINX, SSL and DB connections, diagnosis by rules and an LLM, allow-listed runbook actions, then re-verification. Risky actions go to a person on Telegram (MAPE-K loop, MTTD and MTTR tracked).
- Built SEEANN OFFICE (Next.js 15, NestJS 11, Prisma): read-only access to the production DB, column-level permissions, an append-only audit trail and GHCR CI/CD. In production since Jun 2026.

NemoAI Seoul Feb 2026 – Mar 2026

Full-Stack and AI Developer (left due to a long commute)

- Built an air-gapped RAG platform for public-sector clients: RAGFlow, Milvus, Qwen 80B and OpenWebUI with MCP.
- Built STT, TTS and OCR APIs in FastAPI, tested and benchmarked LLMs, and set up Docker Compose deployment with GitLab CI/CD.

SaM Solutions Germany, remote Sep 2023 – Aug 2025

Backend Developer

- Designed and built MERN apps in TypeScript: Node.js/Express REST APIs with MongoDB (Mongoose) models, and the React screens that use them.
- Implemented authentication with JWT, Express-Session and bcrypt: login, session handling and protected routes on both server and client.
- Configured DNS and firewalls on Linux servers, deployed with Docker and NGINX, automated releases with GitHub Actions and PM2, and monitored the sites with Grafana.

SDLC Corp Poland, remote Aug 2022 – Sep 2023

Backend Developer

- Developed E-CIP, a React content app for a US library information service, and customised the product for the US market.
- Worked on a real-time MARC solution for the library industry: the Node.js backend that turns submitted book data into MARC records.
- On the frontend, built React screens with Redux and Axios for entering and reviewing book data.

Incode Group Kraków, Poland, remote Sep 2020 – May 2022

Frontend Developer

- Built web app screens and reusable components in React and TypeScript, with Redux for state and Axios for REST APIs.
- Implemented responsive layouts and animations (Framer Motion, Anime.js) and page routing with React Router.

PROJECTS

Point2Point Taxi service for Uzbekistan · personal project, MVP

Rider and driver apps (Expo) and an ops panel (Next.js) on a NestJS + Fastify modular monolith with PostgreSQL, Redis, BullMQ and self-hosted OSRM. State change, timeline and outbox in one transaction, a dispatch saga with 15 s offers, a double-entry ledger in tiyin. 119 unit tests.

Milli National messenger for Uzbekistan · personal project, prototype

Per-user update log (pts and getDifference), idempotent sends with ScyllaDB LWT, NATS JetStream, a uWebSockets.js gateway, connection draining on deploy.

Highland.uz Property platform · side project

Next.js, NestJS GraphQL, MongoDB, deployed with Docker on a VPS.

SKILLS

Frontend	Next.js (App Router, SSR, RSC), React, TypeScript, Tailwind CSS, shadcn/ui, Radix UI, TanStack Query, TanStack Table, Zustand, Motion, Sonner, Web Push, i18n, SvelteKit, Univer
Backend	Node.js, NestJS, Express, Fastify, Python (FastAPI), Socket.IO, SSE, REST, GraphQL, protobuf, zod, uWebSockets.js
Data	MySQL, PostgreSQL, MongoDB, Redis, ScyllaDB, primary/replica, Prisma, Kysely, Drizzle, BullMQ, NATS JetStream, Milvus
Infra	AWS (EC2, S3), Terraform, Docker, NGINX, load balancing, PM2, GitHub Actions, GitLab CI, Certbot, Prometheus, Grafana, Linux, Windows
Scale · Ops	load testing, bottleneck analysis, horizontal scaling, caching, connection budgets, cost optimization, self-healing ops agents (health checks, runbooks, Telegram alerts)
Tooling	Turborepo, Turbopack, pnpm workspaces, ESLint (FSD rules), Jest, Vitest, Playwright
3D · AI	vtk.js, Three.js, WebGL2, Web Workers, DICOM / VTI / STL, nnU-Net pipelines, NumPy, RAG (RAGFlow, Milvus, Qwen), LLM tool calling, MCP
Mobile	React Native, Expo, Expo Router, NativeWind, MapLibre, SQLite

EDUCATION

Jeonbuk National University · Bachelor's degree · 2017 – 2024

LANGUAGES

Korean (fluent) · English (fluent) · Russian (fluent) · Uzbek (native)